AI ACADEMY · LEVEL 1 · GRADE 6 · WEEK 20 · STUDENT LESSON

Recommendations

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Helpful Classroom Sorter

THE CONNECTED WEEK

Check requirements. Rank preferences. Audit feedback.

Trace a book recommendation, change an activity requirement and inspect the limits of clicks and missing exposure.

Keep one habit: Verify eligibility → score only eligible options → preserve the event and its limits.

MAIN LESSON · SECTION 1

Which options may be recommended at all?

> Keep requirements, preference points and later feedback in separate stages.

Start with a high-scoring excluded book

A book can match every preference and still fail an essential requirement. Decide eligibility first, then rank the verified eligible options and explain the remaining tradeoff.

Retrieve the language requirement

A book has the highest similarity score but is in a language the reader cannot use. Should the score override the language requirement? Explain.

Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the **evidence** shows was wrong.

By the end, you can...

- I can apply **eligibility** requirements before **ranking** candidates by **preference**s.
- I can explain a **recommendation** using its actual scoring rule and limits.
- I can interpret feedback in the context of what was shown and distinguish clicks from satisfaction.

▼ **Earlier wording, alternatives and picture transcripts**

Your learning target

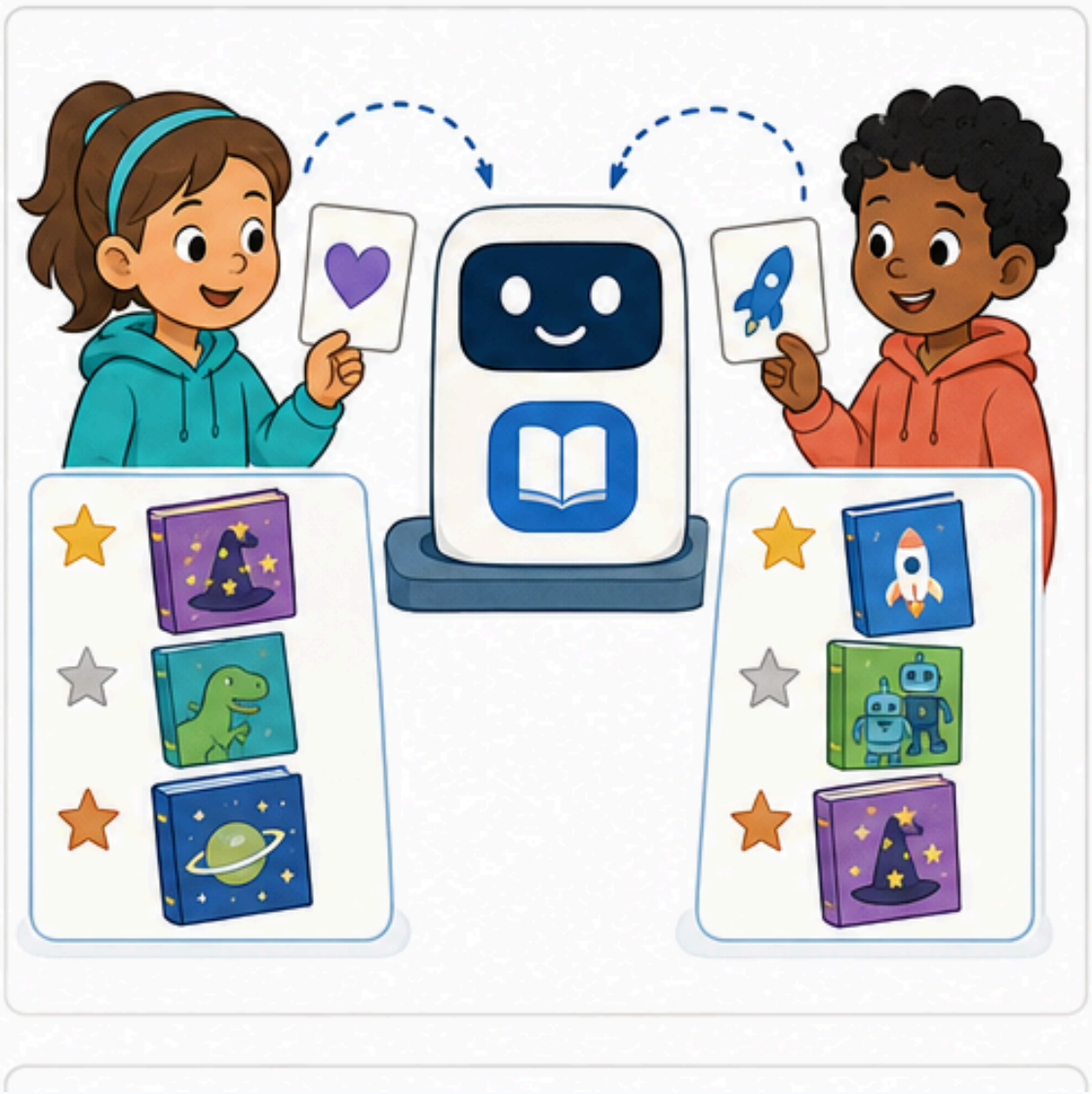
Remember before reading

Without opening the old lesson, explain one key idea from Similarity. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from **Prediction** Is Not Fact. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Build a Human Classifier. Give an example and a mistake that the idea helps you avoid.

A recommender ranks possibilities using goals, information, and **tradeoff**s.



LEARNING TARGET A system shows only adventure books and records only adventure clicks. It concludes the reader dislikes poetry. Explain the missing **evidence** and propose a small test.

Learner name:

Readiness for the connected explanation

This week’s end goal

By the end, I can filter requirements before **ranking preference**s and avoid overinterpreting clicks.

MAIN LESSON · SECTION 2

Filter requirements before ranking preferences

> BookRec1 serves the stated fictional reader using a complete candidate list and tie policy.

Separate required and preferred properties

A **recommendation** suggests an option for a particular purpose. Start with requirements: conditions every suggested item must meet. Then use **preference**s to rank the eligible options.

An item that fails a requirement cannot make up for it by collecting more **preference** points.

Read BookRec1's rule

A fictional reader needs an available English book. The reader prefers topic space and format comic. BookRec1 gives eligible books 2 points for space and 1 for comic; a known mismatch gives 0.

It returns every eligible book tied for the highest score. If none are eligible, it reports no eligible option rather than ignoring a requirement.

Read the four supplied book records

The candidates are J: space, prose, English, available; K: animals, comic, English, available; L: space, comic, Spanish, available; M: space, comic, English, unavailable. These are invented records.

The purpose is to trace a **recommendation**, not to infer anything about real students or collect their personal **preference**s.

Explain the two stages

- List BookRec1's two requirements and two weighted preferences. Which questions are answered by filtering, and which are answered only after comparing the eligible scores?

▼ Earlier wording, alternatives and picture transcripts

PICTURE STORY

Look Closely Before You Explain

Pictures give clues; **evidence** tells us what the clues mean.



MYSTERY Two readers can receive different book **ranking**s because their goals, reading level, interests, and novelty **preference**s differ.

What do you notice?

Name three visible details without explaining them yet.

Circle the detail most connected to the system's decision.

Name one detail that may be decoration or distraction.

Make a first **prediction** and state the clue that supports it.

My **observation**s, **prediction**, and **evidence**:

GOOD THINKING A **prediction** can change when stronger **evidence** appears.

A closer explanation

Explain the actual reason for the recommendation

> Mention the preference the chosen item does not match as well as the matches that earned its points.

Read eligibility and scores

Requirements come before preference scores

CANDIDATE	ELIGIBILITY CHECK	RESULT
J: space and prose	English; available	Eligible; 2 points
K: animals and comic	English; available	Eligible; 1 point
L and M	Language or stock fails	Excluded from ranking

BookRec1 recommends J from this candidate set. L and M match both preferences but fail a requirement. The recommendation does not guarantee that the reader will enjoy J.

Trace exclusions before ranking

J passes both requirements and scores 2. K passes and scores 1. L fails the English requirement; M fails availability. Do not place L or M above J because they would earn 3 **preference** points.

They are excluded before the eligible scores are compared.

Write a faithful explanation for J

A faithful explanation says J is available in English and has the highest eligible score because its space topic matches the stated **preference**. It should also mention that J is prose, not comic.

Claiming that the system knows the reader will love J would go beyond the computation and the **evidence**.

Explain one deciding stage

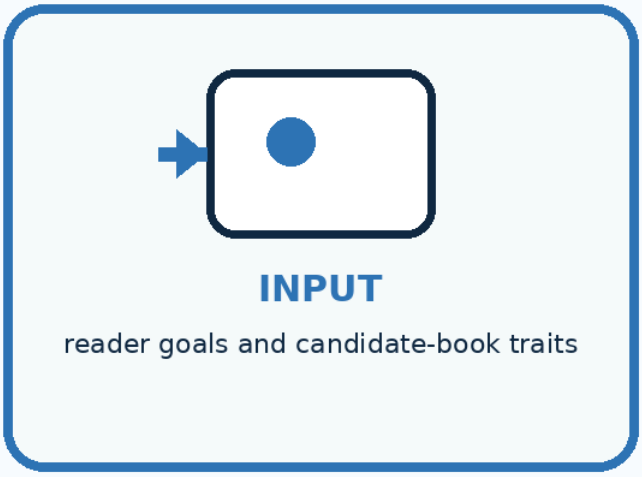
- Point to the **input**, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

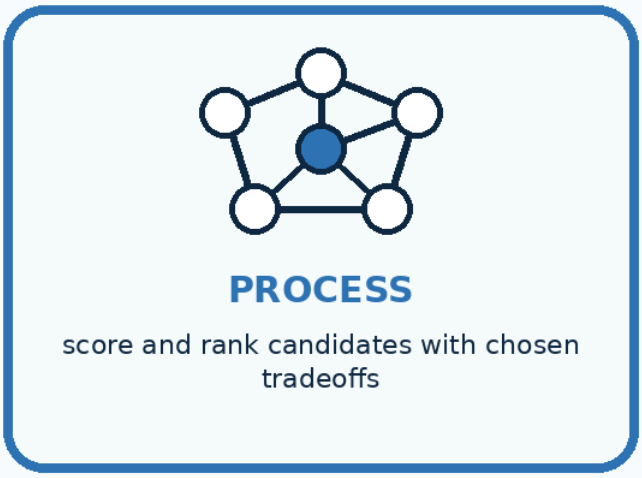
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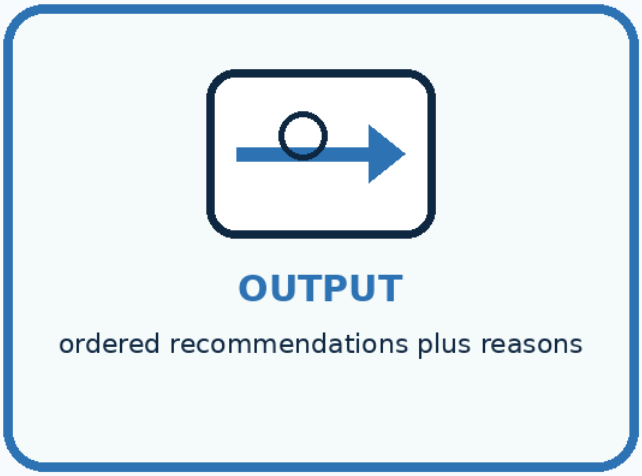
VISUAL **MECHANISM**

Input → Process → Output

Follow the information instead of imagining magic inside the machine.







INPUT reader goals and candidate-book traits

PROCESS score and rank candidates with chosen **tradeoff**s

OUTPUT ordered **recommendation**s plus reasons

CORE IDEA A recommender ranks possibilities using goals, information, and **tradeoff**s.

UNDER THE HOOD **Recommendation** is **ranking**, not merely classification. An **objective** combines relevance signals and constraints; **tradeoff**s may include familiarity, **diversity**, novelty, safety, and availability.

The top score reflects the **objective**, not universal quality.

Draw a fresh example with all three stages:

A closer explanation

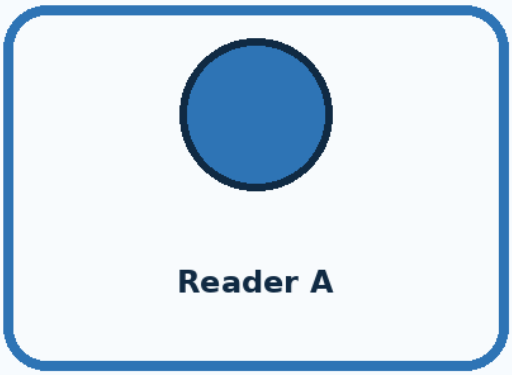
Connect the picture and the explanation

MAIN LESSON · SECTION 4

Compare recommendations and feedback clues

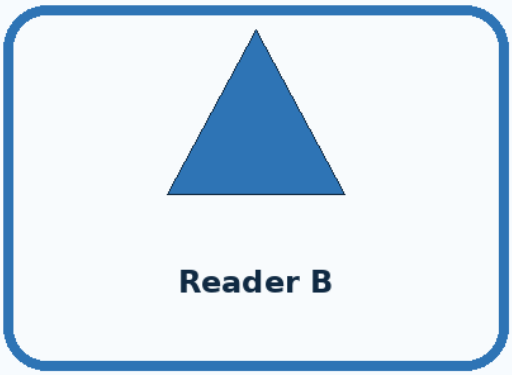
> Use the stated goal and recorded exposure before interpreting a suggested item or click.

Pictures paired with their evidence



The case to investigate

CASE 1 — Reader A Clue: likes survival and long books. Meaning: rank adventure higher.



The case to investigate

CASE 2 — Reader B Clue: wants short humor. Meaning: rank comedy higher.



The case to investigate

CASE 3 — Same score Clue: two books tie. Meaning: need tie policy.



The case to investigate

CASE 4 — Unavailable book Clue: high relevance but unusable. Meaning: constraint changes rank.

Discuss the confusing pair

- Which two pictures are easiest to confuse? What **evidence** separates them?

Keep the limit in view

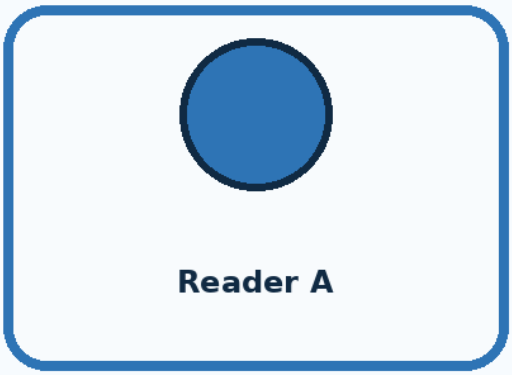
BOUNDARY A **recommendation** optimizes chosen goals and data; it does not discover what a person truly wants.

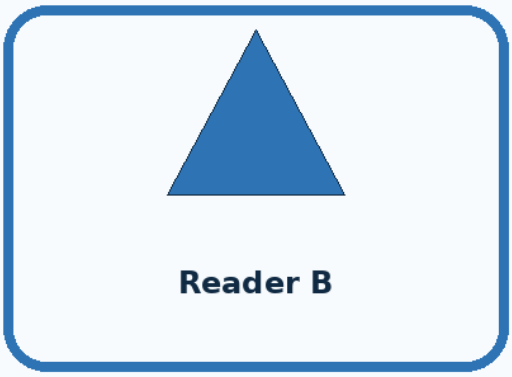
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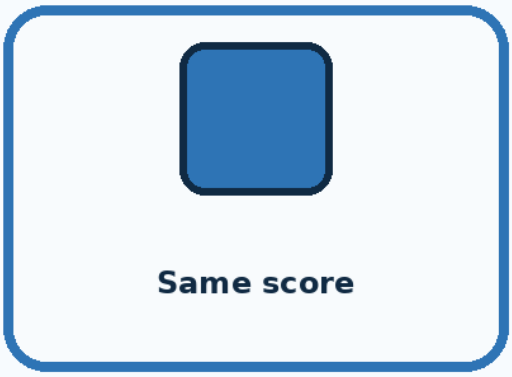
PICTURE GALLERY

Four Cases, Four Clues

Use the visual cue, then read the **evidence** carefully.









CASE 1 — Reader A Clue: likes survival and long books. Meaning: rank adventure higher.

CASE 2 — Reader B Clue: wants short humor. Meaning: rank comedy higher.

CASE 3 — Same score Clue: two books tie. Meaning: need tie policy.

CASE 4 — Unavailable book Clue: high relevance but unusable. Meaning: constraint changes rank.

Change one thing in the pictured case

Change: Change the maximum time to 10 minutes.

Prediction: Only C remains eligible and is recommended.

Reason: B takes **12 minutes**; A is still noisy.

MAIN LESSON · SECTION 6

A changed requirement can change the recommendation

> This separate break chooser first requires quiet and at most 12 minutes, then scores puzzle and solo preferences.

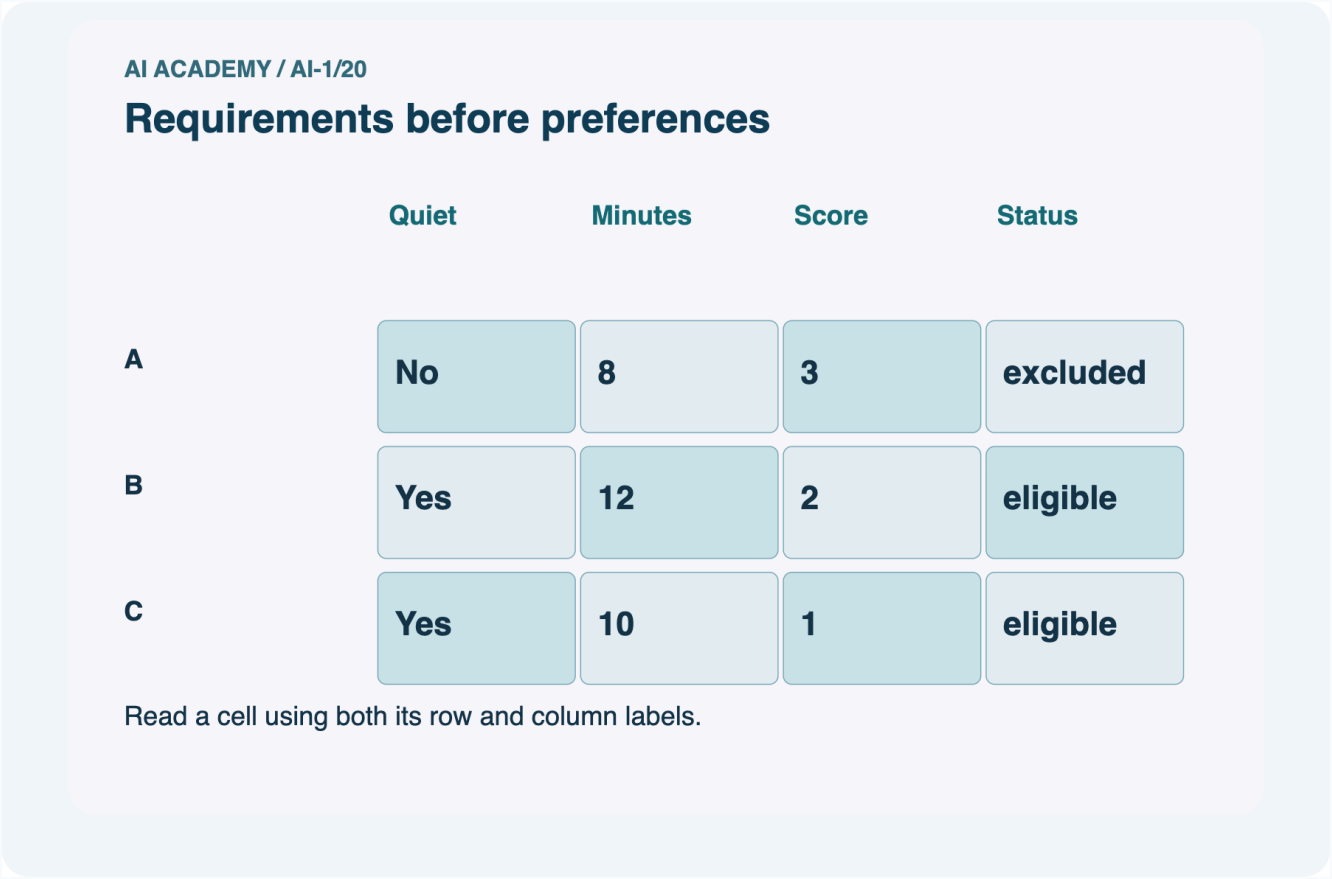
The activity policy and supplied candidates

A fictional classroom break chooser recommends a suitable paper activity.

Must be quiet and take at most **12 minutes**. Score 2 for puzzles and 1 for solo.

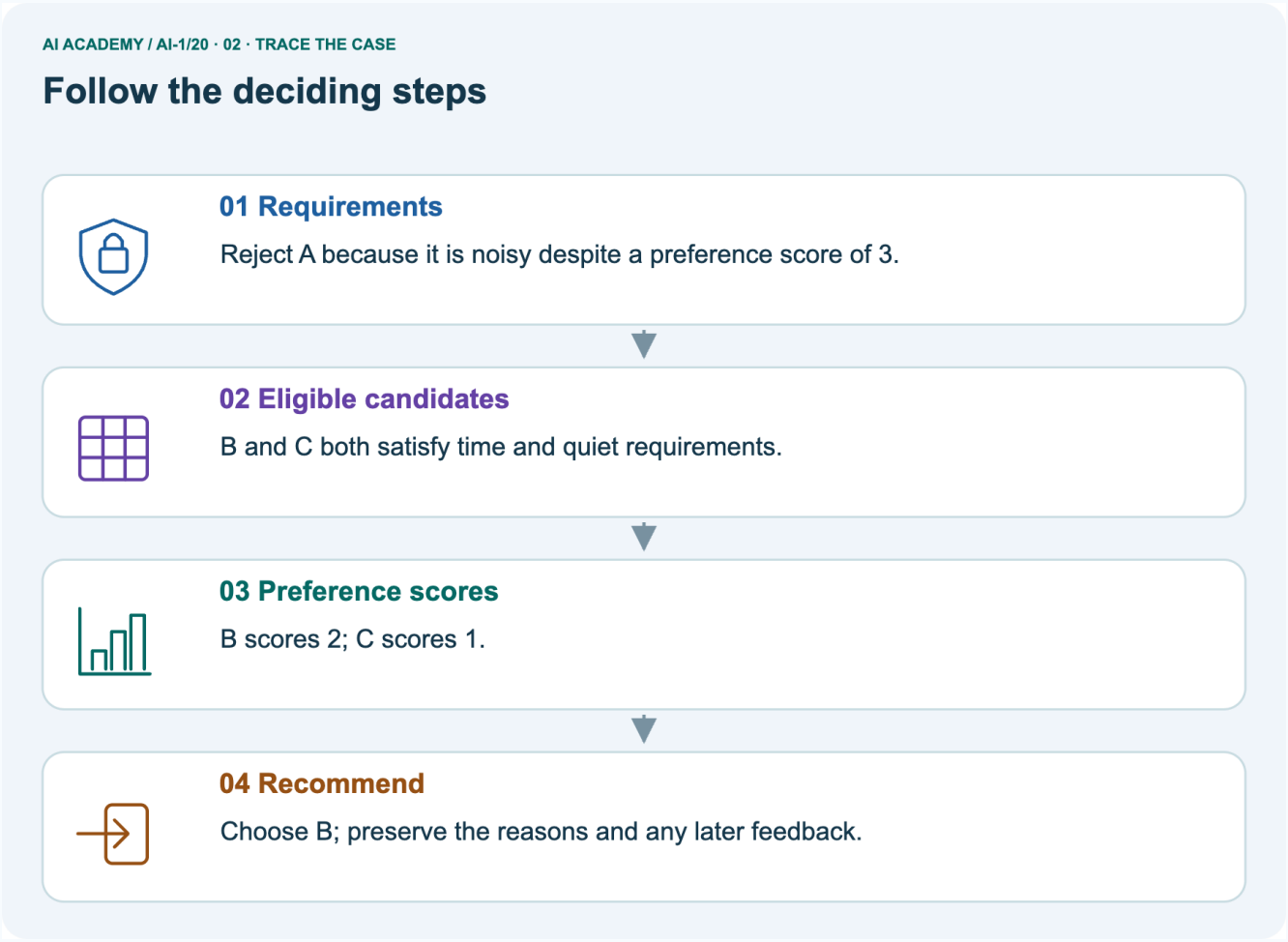
A puzzle,solo,noisy,8 min; B puzzle,group,quiet,12 min; C drawing,solo,quiet,10 min.

Compare requirements and preference matches



Follow the labelled picture with the matching explanation. Open the vector diagram

Trace eligibility before ranking



Follow the labelled picture with the matching explanation. Open the vector diagram

Explain exclusion and the limits of a click

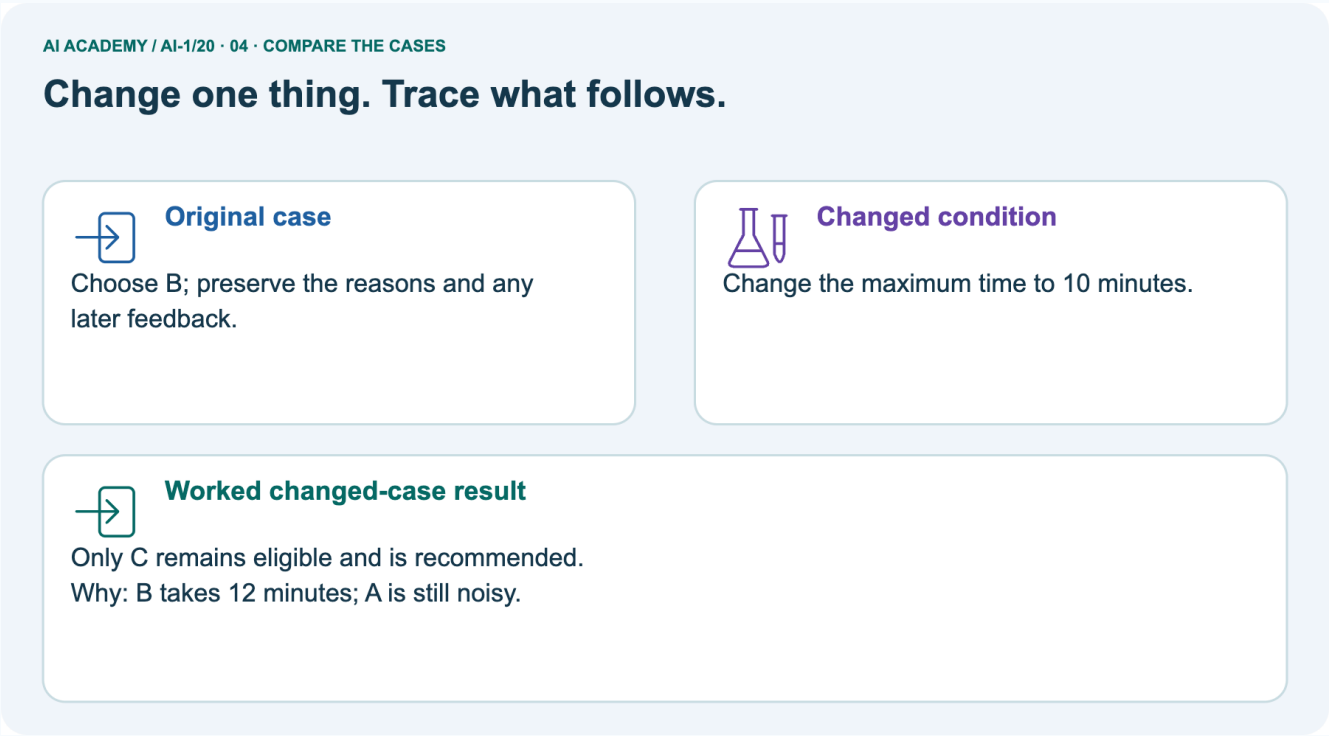
A hard requirement determines whether an option can be chosen at all. **Preference**s rank the eligible options. The apparent best **preference** match, A, is not suitable under the quiet requirement.

If a learner clicks C instead, the event alone does not tell us whether they disliked B, misclicked or wanted variety.

Predict with a tighter time limit

- Before checking, change only the activity time limit from 12 to 10 minutes. Keep quiet as a requirement and keep the preference weights unchanged. Filter again, then score the remaining eligible options.

▼ Check the new eligible choice



Follow the labelled picture with the matching explanation. Open the vector diagram

Keep satisfaction separate from an interaction

One click is an observed event, not proof of satisfaction or a stable personal **preference**.

Explain the unchanged preferences

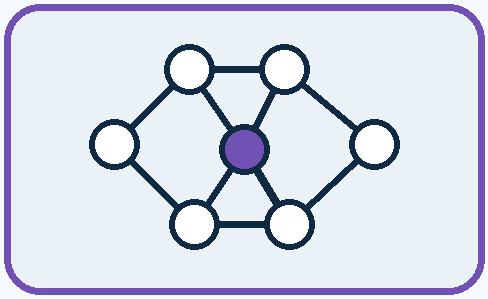
- Read the labelled picture: Why does the **recommendation** change without changing anyone's **preference**s?

▼ Earlier wording, alternatives and picture transcripts

WORKED EXAMPLE

Watch the Reasoning, Not Just the Answer

Every step connects a claim to observable **evidence**.



CASE Two readers can receive different book **ranking**s because their goals, reading level, interests, and novelty **preference**s differ.

STEP 1 — OBSERVE reader goals and candidate-book traits

STEP 2 — TRACE score and rank candidates with chosen **tradeoff**s

STEP 3 — CONCLUDE ordered **recommendation**s plus reasons

MODEL REASONING Use the same transparent scoring method with each reader's stated **preference**s. Different **input**s create different totals. Report **tradeoff**s, ties, and one option outside the top rank to preserve choice.

DO NOT OVERCLAIM A **recommendation** optimizes chosen goals and data; it does not discover what a person truly wants.

What new **evidence** could change this answer?

A closer explanation

Recorded event	What it establishes	What it does not establish
J was shown	There was a recorded exposure	The reader noticed or liked it
J was clicked	A recorded interaction occurred	The reader enjoyed the book
Reader said J was too long	A stated reason in this context	All space books are unwanted
K was never shown	No exposure recorded for K	The reader rejected K

A click can reflect interest, curiosity or a mistaken tap. No click can reflect a missed display, lack of time or a decision not to choose it. Preserve the observed event separately from any interpretation.

More direct feedback may clarify a reason, but one response still describes one person and occasion.

Useful feedback records name the item, what was shown, the system version, the event and any stated reason. Use only information needed for the task.

Our paper exercises use invented feedback; they do not require accounts or recording classmates' browsing behaviour.

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

A pictorial worked case: Recommendations

The facts we will use

AI ACADEMY / AI-1/20

Requirements before preferences

	Quiet	Minutes	Score	Status
A	No	8	3	excluded
B	Yes	12	2	eligible
C	Yes	10	1	eligible

Read a cell using both its row and column labels.

Follow the labelled picture with the matching explanation. Open the vector diagram

AI ACADEMY / AI-1/20

Recommendations

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Requirements

Reject A because it is noisy despite a preference score of 3.



2. Eligible candidates

B and C both satisfy time and quiet requirements.



3. Preference scores

B scores 2; C scores 1.



4. Recommend

Choose B; preserve the reasons and any later feedback.

CHANGE ONE THING

Change the maximum time to 10 minutes.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Requirements	Reject A because it is noisy despite a preference score of 3.
2. Eligible candidates	B and C both satisfy time and quiet requirements.
3. Preference scores	B scores 2; C scores 1.
4. Recommend	Choose B; preserve the reasons and any later feedback.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

What this example does not establish

Change one thing: Change the maximum time to 10 minutes. Predict the result before checking the explanation. What remains the same?

Use feedback to ask a better question

> Keep required information verified and avoid turning one comment into a permanent preference.

Inspect what “too long” could mean

If feedback says J is too long, length may be a relevant missing feature. Ask what limit matters before adding a requirement.

Do not automatically remove all space books or silently turn one comment into a permanent **preference**. Keep the original feedback and document any change to the **recommendation** rules.

Retain pending candidates and a no-verified-option outcome

Check a revised version on reserved paper cases, including ties, no eligible choices and missing required information.

If a required fact is unknown, mark that candidate pending verification and report the limitation; do not assume it passes. If all candidates are pending or excluded, return no verified eligible option.

A **recommendation** leaves the final choice with the reader.

Explain the display-and-click feedback loop

- CLAIM A The highest-ranked item is best for everyone.
- CLAIM B Personalization removes the need for user choice.

STRONG ANSWER FRAME I claim _____ because the **evidence** shows _____. The **process** is _____. One limit is _____.

A system recommends only the books it showed last week because those are the only books with clicks. Explain how its own display choices could keep other books out of consideration and name one improvement to the evidence collection

▼ Earlier wording, alternatives and picture transcripts

COMPARE

Same Result, Different Reason

A comparison helps reveal the true concept boundary.

Case	Evidence	What it shows
Reader A	likes survival and long books	rank adventure higher
Reader B	wants short humor	rank comedy higher
Same score	two books tie	need tie policy
Unavailable book	high relevance but unusable	constraint changes rank

Repair two tempting claims

Repair:

Repair:

A closer explanation

Practise: record what was shown as well as what was clicked

> An item with no recorded exposure has not thereby been rejected.

Trace how the system shapes its later evidence

Suppose a system shows only space books, then collects clicks only on those books. If it treats those clicks as proof that space is the only wanted topic, it may keep showing space.

Its choices helped produce the **evidence** it later uses. The absence of animal-book clicks says little when animal books were never shown.

> **A log says J was shown and clicked; K was never shown. A designer wants to label J enjoyed and K disliked. Write the two defensible records, identify the unsupported labels, and propose a useful way to ask for feedback without assuming the answer**

Gather optional stated feedback without assuming its answer

A better investigation can ask the fictional reader directly or offer a small set of eligible alternatives with clear choice and optional feedback. Record what was offered as well as the response.

Such a comparison still needs care: order, context and missing responses can affect what is observed. Do not call an unseen item rejected.

Use W2 to explain J's recommendation honestly

Use workbook W2 for the connected example "**Recommendation**s and feedback". First fill the **input** and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan	Do	Check
What is given? Which rule or procedure applies?	Record each intermediate step.	Does the result follow? What remains unknown?

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

Challenge cancelling a failed requirement with points

- Use the worked case for **Recommendation**s and feedback. Proposed explanation: “A high score can cancel a failed requirement”. Decide whether this explanation is supported.

  Point to the step or supplied fact that decides; explain your repair if needed.

- Answer on your own before discussion. If you are unsure, say which step you need help with.

Return to the changed activity limit

Why does the **recommendation** change without changing anyone’s **preference**s?

> **My explanation using a pictured step and the given facts**

This is supported practice. Keep the separately named independent case for an attempt before feedback.

▼ **Earlier wording, alternatives and picture transcripts**

GUIDED PRACTICE

Try It, Check It, Explain It

Use support now so you can succeed independently later.



Explain how the picture story demonstrates this idea: A recommender ranks possibilities using goals, information, and **tradeoff**s.

Use a fresh example to explain **ranking**.

Trace a new **input** → **process** → **output** example.

Find and repair the error in: Personalization removes the need for user choice.

Invent one edge case and name the safe response.

PARTNER CHECK Point to the **evidence**. Do not accept an answer that only repeats the vocabulary word.

A closer explanation

Practise with less help: Explain a recommendation honestly

Check your reasoning

Explain the picture

MAIN LESSON · SECTION 9

Investigate a two-reader book ranker

> Use explicit fictional requirements and priorities, then inspect missing eligibility and exposure feedback.

The records, two profiles and ranking policy

Use fictional books U=mystery,prose,English,available; V=animals,comic,English,available; W=mystery,comic,Spanish,available; X=animals,prose,English,unavailable. Both readers require available English books. Reader R1 scores mystery 2 and comic 1. Reader S1 scores animals 2 and prose 1. Known mismatches score 0; return all tied highest verified eligible books. Unknown required facts mark a candidate pending, not verified.

Predict, filter, rank and interpret feedback

1. Predict eligibility and scores for each reader before tracing with a partner. Explain why their recommendations differ. For a second supplied case, make U and V's availability unknown while retaining W and X's records; state the resulting honest response. A feedback card says only R1 was shown U and clicked it; V was not shown. Record the event without labelling U enjoyed or V disliked.

2. Keep a normal case and an edge case clearly separated.

3. Record what happened—even when it does not match your **prediction**.

4. Explain the result, one limitation, and the next useful test.

Keep policy versions and observed events

Case	Prediction	Observed result	What it means
Normal			
Edge			

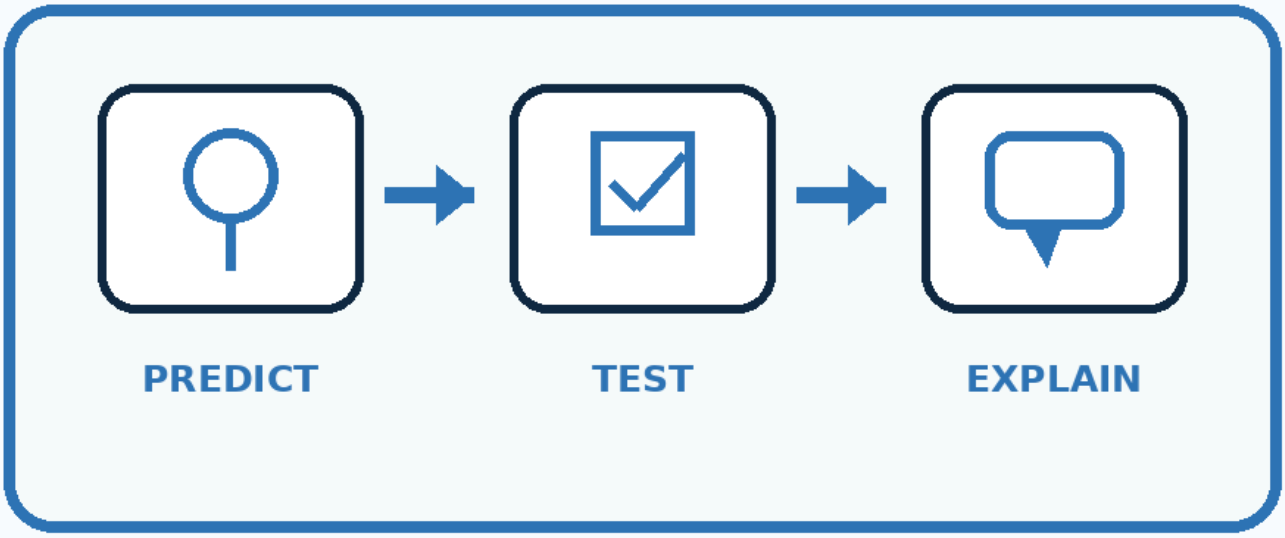
Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep "not run" if you only traced the procedure.

▼ Earlier wording, alternatives and picture transcripts

EXPLORER LAB

Two-Reader Book Ranker

Predict first, test honestly, and explain what the **evidence** means.



State your **prediction** before the result is known.

Complete the task: Score books for two readers; explain why **ranking**s differ.

INDEPENDENCE After support is removed, repeat the proof with a fresh case.

Plan → try → check

MAIN LESSON · SECTION 14

Try alone: recommend from the new activity records

> Apply the indoor and at-most-20-minute requirements before preferences, retaining every tied highest scorer.

The new task, records and exposure feedback

A fictional learner needs an indoor activity lasting at most 20 minutes. **Preference**s score 2 for drawing and 1 for solo.

Candidates: A drawing, solo, outdoor, 10 minutes; B drawing, group, indoor, 20 minutes; C music, solo, indoor, 15 minutes; D drawing, group, indoor, 18 minutes. Filter, score and return all highest scorers.

Only B is then shown and selected; C and D are not shown. State what the feedback establishes and what remains unknown.

> **My independent explanation, trace and limits**

Keep one first attempt

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Your teacher has the independent answer and scoring criteria. Complete the matching workbook week to keep your practice **evidence** together.

Use feedback afterwards

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed.

A later check uses a changed case, not a copied answer.

▼ Earlier wording, alternatives and picture transcripts

Independent case: Transfer the process to activities

Week 20 Independent application

Close the worked explanation. Apply the idea to this different question without copying.

A recommender suggests a space book because you liked two space books. Does the suggestion prove you will enjoy it? What feedback could help next time?

My answer, with a trace, calculation or supporting **evidence**

> **Write or draw your reasoning and evidence.**

Create and test

Change one relevant detail. Predict the result before testing. Include a boundary or missing-information case when it helps reveal a limit.

My changed **input** and expected result

> **Write or draw your reasoning and evidence.**

Connect to your project

If the sorter ranks bins, state the **objective** and show why the top suggestion is not a command.

The **evidence** I will keep

Write or draw your reasoning and evidence.

Transfer practice from the original programme

A system shows only adventure books and records only adventure clicks. It concludes the reader dislikes poetry. Explain the missing **evidence** and propose a small test.

Explain the mechanism, evidence and one limit.

Use feedback to improve

MAIN LESSON · SECTION 11

Explain why a sorter suggestion was eligible

> Keep requirements, preferences and observed feedback distinguishable in the project record.

This week’s project decision

THIS WEEK'S CHECKPOINT If the sorter ranks bins, state the **objective** and show why the top suggestion is not a command.

Separate sorter requirements, **preference**s and observed feedback; document why a suggestion was eligible.

Plan filters, scores and review

- What real classroom need does this serve?
- What information is necessary—and what should stay private?
- What happens inside the **process**?
- What normal and edge cases will you test?
- When should it say not sure or ask a person?

Keep the objective and data choices visible

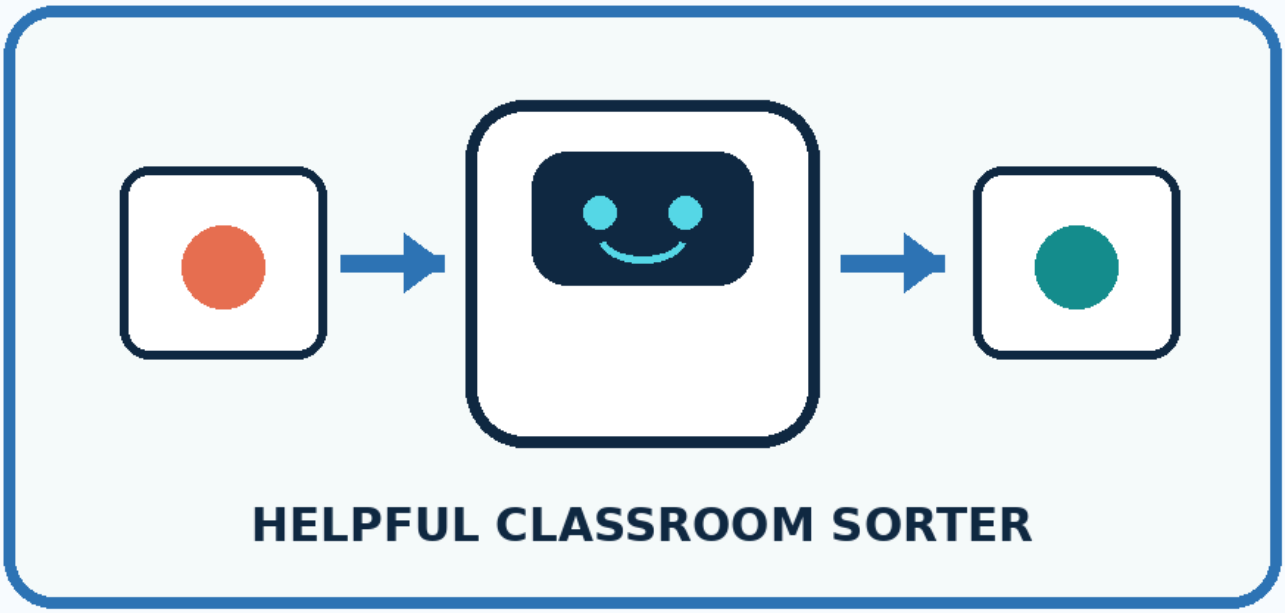
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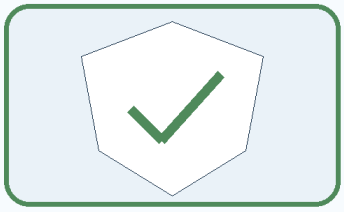
▼ Earlier wording, alternatives and picture transcripts

PROJECT

Helpful Classroom Sorter

Build one coherent system across the year, one **evidence**-based decision at a time.





My project decision:

Evidence for your project

MAIN LESSON · SECTION 16

Finish: justify eligibility, ranking and feedback limits

> Close your notes and explain the two-stage choice and what an observed click cannot settle.

Your exit explanation

- Close your notes. Explain this goal in your own words: Apply **eligibility** requirements before **ranking** candidates by **preference**s

My deciding step, evidence and one remaining question

Choose support and revisit later

If you are unsure, return to the worked picture and explain one arrow. If you are ready, change an **input** or assumption and justify your **prediction**.

Save your workbook **evidence**. At the next review, try a different case before opening your old answer.

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Your quick word guide**

REFERENCE · SECTION 5

Your quick word guide

> Use the words to explain a particular step, record or limit.

Core vocabulary

Word	Meaning	My example
ranking	ordered list from higher to lower priority	
objective	goal the scoring process tries to satisfy	
tradeoff	gain in one goal that may reduce another	
diversity	meaningful variety among recommended options	

Terms for the connected explanation

Word	Meaning
Eligibility	Whether an item meets the requirements for consideration
Preference	A desired feature used to rank eligible choices

Word	Meaning
Recommendation	A suggested option produced under a stated process
Exposure	An opportunity for a person to see an item
Feedback loop	A process in which earlier recommendations influence the feedback used for later recommendations

Repair a claim using a word

- CLAIM TO REPAIR The highest-ranked item is best for everyone.
- Use one vocabulary word and one piece of **evidence** to repair it:

▼ Earlier wording, alternatives and picture transcripts

VOCABULARY

Words You Can See and Use

The badge is a memory cue; the definition does the real thinking work.









Misconception detective

WORD POWER A strong definition helps you decide something in a new case.

Words used in the closer explanation

▼ Choose a stronger thinking challenge

REFERENCE · SECTION 10

Choose a stronger thinking challenge

> Choose the next useful challenge rather than treating the whole ladder as compulsory.

Pick your next step

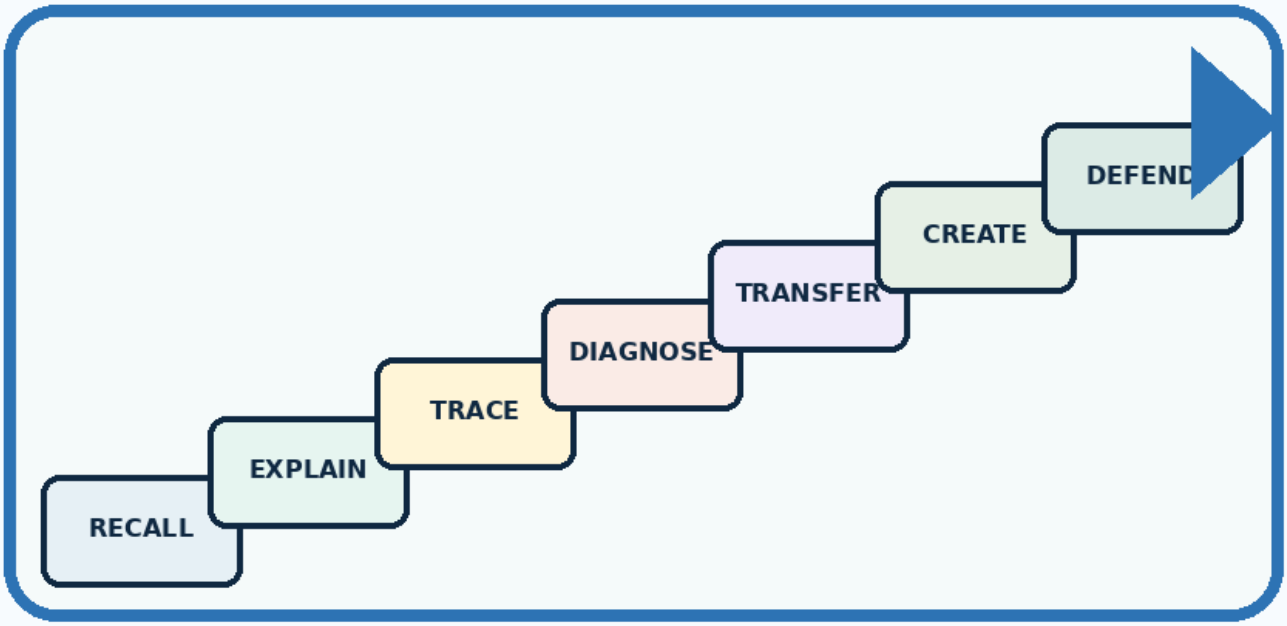
- RECALL Define **ranking**.
- EXPLAIN A recommender ranks possibilities using goals, information, and **tradeoff**s.
- TRACE Show the **input**, **process**, and **output**.
- DIAGNOSE Repair: The highest-ranked item is best for everyone.
- TRANSFER Apply the idea to a new home or classroom case.
- CREATE Design one revealing test.
- DEFEND Give a claim, **evidence**, **mechanism**, limit, and safe action.

▼ Earlier wording, alternatives and picture transcripts

PRACTICE LADDER

Climb from Recall to Defense

Each step asks for a stronger kind of understanding.



RETRY RULE A coached answer becomes mastery only after you solve a different case independently.

▼ **Reference: check your explanation**

REFERENCE · SECTION 12

Reference: check your explanation

> Use after practice. Transfer the process to activities is the selected independent case.

Check the evidence and its limit

- ☐ Name or draw the fresh case.
- ☐ Trace **input**, **process**, and **output**.

- ☐ State the conclusion and strongest **evidence**.
- ☐ Name one limit, failure, missing clue, or fairness/privacy concern.
- ☐ Explain one safe next action or improvement.

Record support

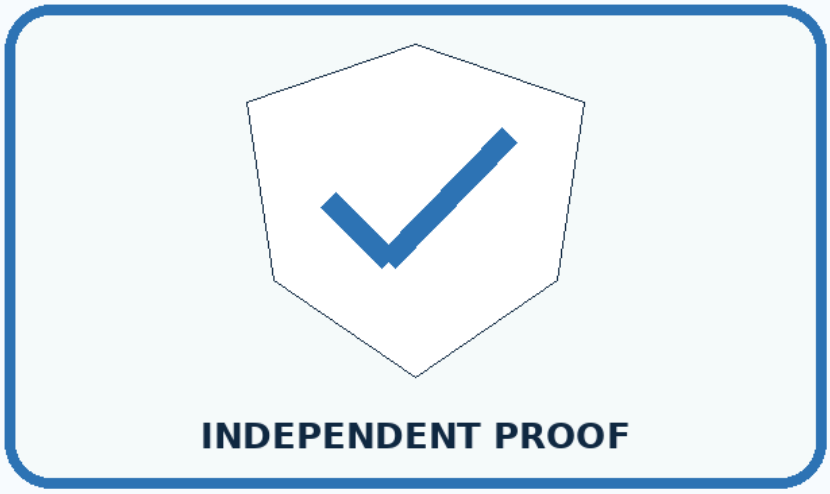
My level: ☐ Not yet ☐ With one hint ☐ Independently ☐ I can teach it

▼ Earlier wording, alternatives and picture transcripts

MASTERY

Checklist for explaining this week's idea

Use after practice. Transfer the process to activities is the selected independent case.



EXPLANATION CHECK Check all requirements before **ranking** preferences, explain the actual score and tradeoff, and keep exposure and interaction separate from satisfaction.

Independent **evidence**:

SELF-CHECK Use the same transparent scoring method with each reader's stated **preference**s. Different **input**s create different totals. Report **tradeoff**s, ties, and one option outside the top rank to preserve choice.

NEXT BRIDGE Carry today's **evidence** habit into: Good Data, Bad Data.

▼ **Extra practice: Recommendations shape future evidence**

REFERENCE · SECTION 13

Extra practice: Recommendations shape future evidence

> Try the question before opening its worked explanation.

Read the supplied case and question

A recommender suggests adventure books after one adventure-book choice.

Explain its **evidence** and limitation.

Your explanation

- Which detail supports the answer, and why does it matter?

> **Write or draw your reasoning and evidence.**

▼ Check after trying

The past choice supports considering adventure, but does not capture every interest. Ask for feedback and offer variety.

Evidence or stage	Value or result
Past choice	One adventure book
Suggestion	More adventure
Feedback	Ask what the reader liked
Next suggestion	Use feedback and offer variety

▼ Earlier wording, alternatives and picture transcripts

Week 20 Worked case

Use the supplied details to practise the weekly concept before attempting the independent question. This case extends the illustrated lesson.

The situation

The question

Worked explanation

Explain it in your own words

Trace the supplied case visually

Recommendations shape future evidence

A recommender suggests adventure books after one adventure-book choice.

Explain its **evidence** and limitation.

The past choice supports considering adventure, but does not capture every interest. Ask for feedback and offer variety.

▼ Reference: original lesson checklist

REFERENCE · SECTION 15

Reference: original lesson checklist

> Use an additional checklist when it helps inspect your explanation.

Targets and evidence

By the end, I can	How I will show it
Apply eligibility requirements before ranking candidates by preferences	Explain the deciding step, show a trace or test, and state one limit.
Explain a recommendation using its actual scoring rule and limits	Explain the deciding step, show a trace or test, and state one limit.
Interpret feedback in the context of what was shown and distinguish clicks from satisfaction	Explain the deciding step, show a trace or test, and state one limit.

Keep your route and first prediction

Remember → watch and explain → try with help → investigate → try independently → keep project **evidence**. Your teacher may spread this lesson across several classes.

Keep your first **prediction**. Compare it with what the **evidence** shows and explain any change.

▼ Earlier wording, alternatives and picture transcripts

Start with this question: A fictional classroom break chooser recommends a suitable paper activity.

▼ Extra practice: pictures, repair and feedback

REFERENCE · SECTION 17

Extra practice: pictures, repair and feedback

> Attempt each question before opening its explanation. Use one browser or paper copy.

First supported question

- A fictional classroom break chooser recommends a suitable paper activity.
- Which labelled step matches this detail? Choose B; preserve the reasons and any later feedback.
- 1. Recommend
- 2. Requirements
- 3. Eligible candidates
- 4. **Preference** scores
- Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Practice 1: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ Check the first explanation

After your attempt, compare with the supported explanation: Recommend Choose B; preserve the reasons and any later feedback. A hard requirement determines whether an option can be chosen at all.

Preferences rank the eligible options. The apparent best **preference** match, A, is not suitable under the quiet requirement.

If a learner clicks C instead, the event alone does not tell us whether they disliked B, misclicked or wanted variety.

Changed-case question

- Changed-case practice: Change the maximum time to 10 minutes. Which conclusion is supported by the supplied facts?
- 1. Only C remains eligible and is recommended.
- 2. This one example guarantees correct results for every future **input**.
- 3. We can decide the result without examining the changed **input** or the procedure.
- Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Practice 2: explain the deciding evidence

> **My selected conclusion, deciding fact and explanation**

▼ Check the changed case

After your attempt, compare with the supported explanation: Only C remains eligible and is recommended. B takes **12 minutes**; A is still noisy.

Repair and save your work

Review Similarity (ai-1/19). Goal: Calculate similarity using explicitly selected features and a stated scoring rule. Short practice: Point to the matches supporting C's new score and explain why there is no tie..

Return to this week and explain what you changed.

Review **Prediction** Is Not Fact (ai-1/17). Goal: Separate a recorded **prediction** from the **observation** used to check it. Short practice: Which field changes the **observation** label, and which field changes the model **prediction**?.

Return to this week and explain what you changed.

Review Build a Human Classifier (ai-1/14). Goal: Construct a classifier with a stated **input** format, selected rule and recorded **output**.

Short practice: Trace 0 cm and explain why clarification is different from predicting short..
Return to this week and explain what you changed.

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

▼ Earlier wording, alternatives and picture transcripts

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Repair a missing building block

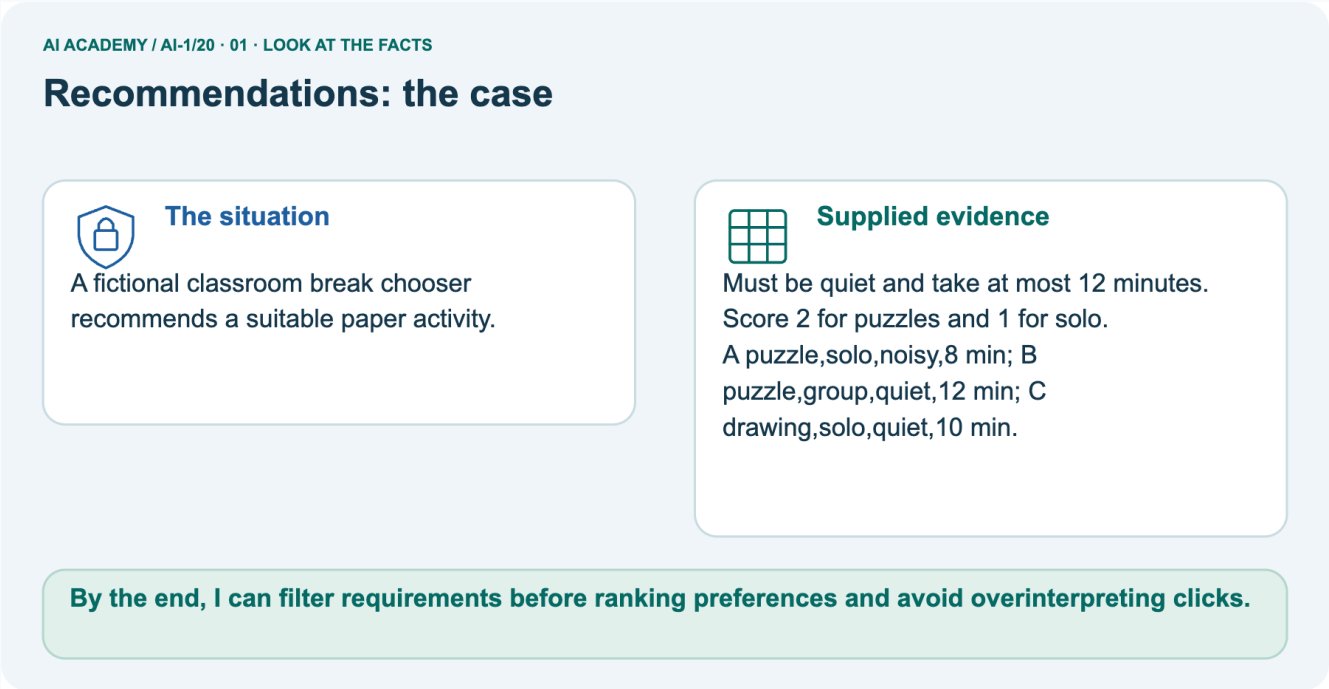
Keep your completed work

▼ Reference: the six-picture journey

Reference: the six-picture journey

> Revisit the picture that serves the current question; key diagrams already sit beside the main explanation.

01. Recommendations: the case



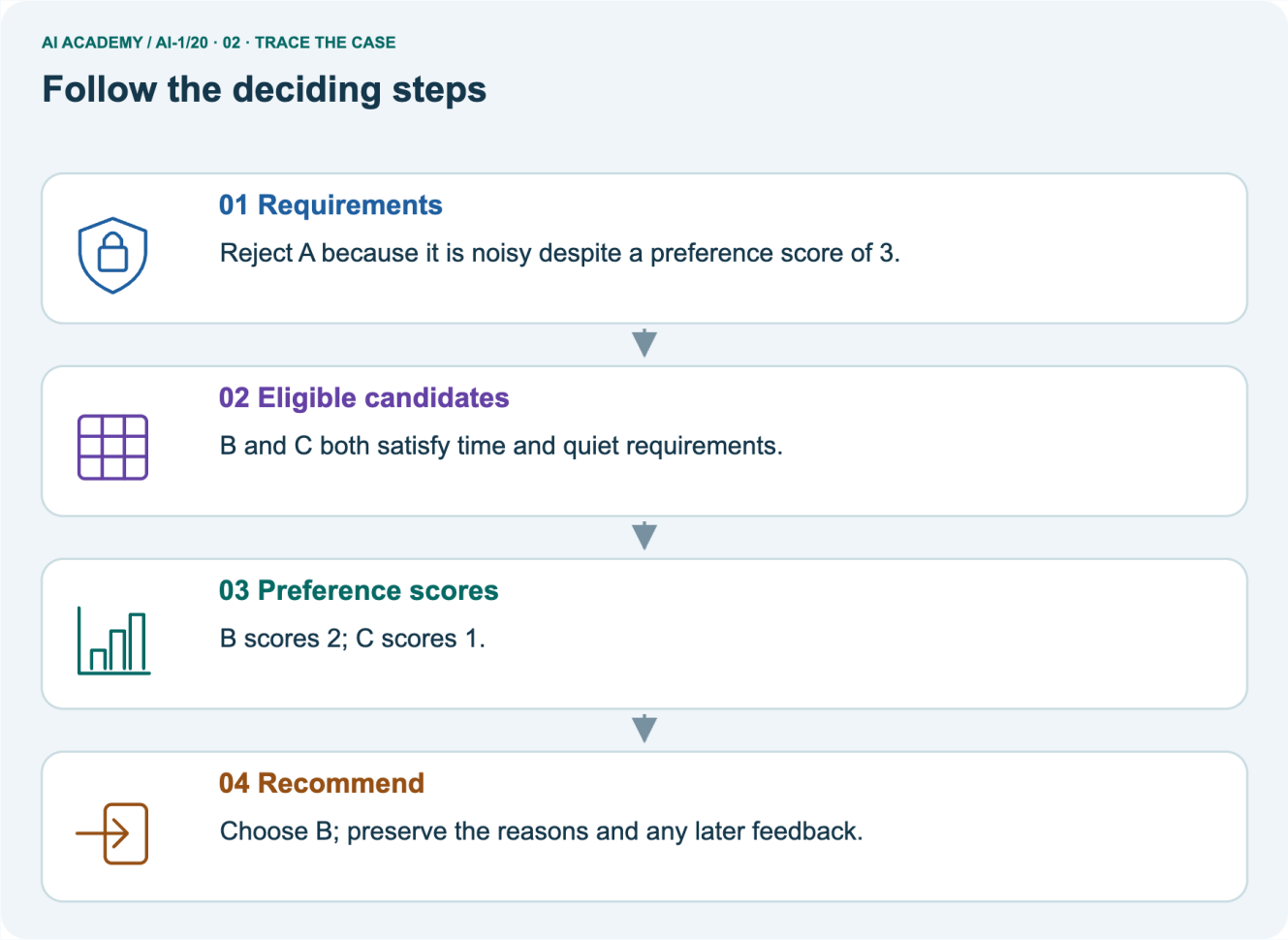
Follow the labelled picture with the matching explanation. Open the vector diagram

A fictional classroom break chooser recommends a suitable paper activity. Supplied facts: Must be quiet and take at most **12 minutes**. Score 2 for puzzles and 1 for solo.

A puzzle,solo,noisy,8 min; B puzzle,group,quiet,12 min; C drawing,solo,quiet,10 min.

Point to the given fact that matters most. Explain why it matters.

02. Follow the deciding steps

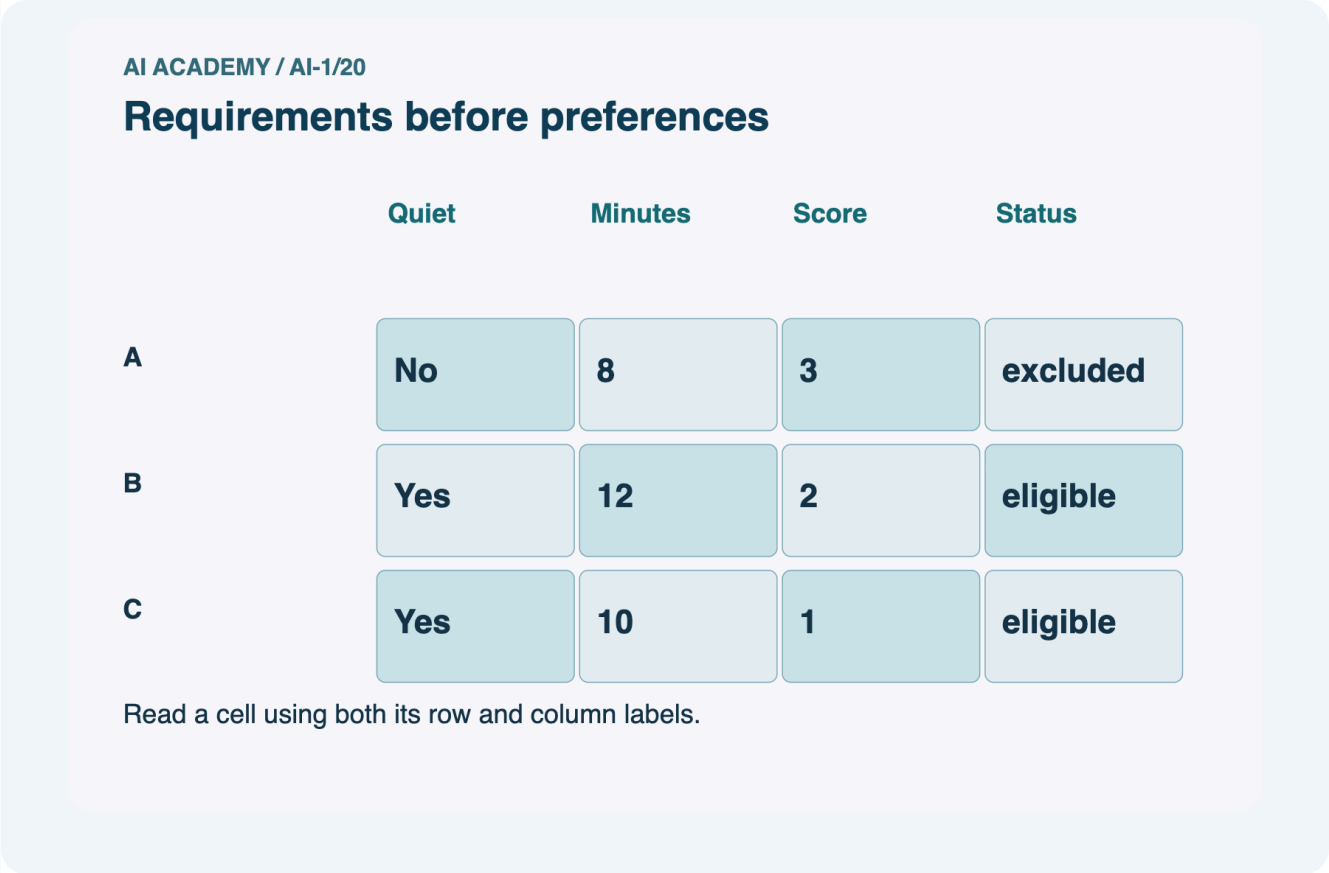


Follow the labelled picture with the matching explanation. Open the vector diagram

Requirements: Reject A because it is noisy despite a **preference** score of 3. → Eligible candidates: B and C both satisfy time and quiet requirements. → **Preference** scores: B scores 2; C scores 1. → Recommend: Choose B; preserve the reasons and any later feedback.

Point to each numbered stage and explain which recorded **input** it uses.

03. Requirements before preferences

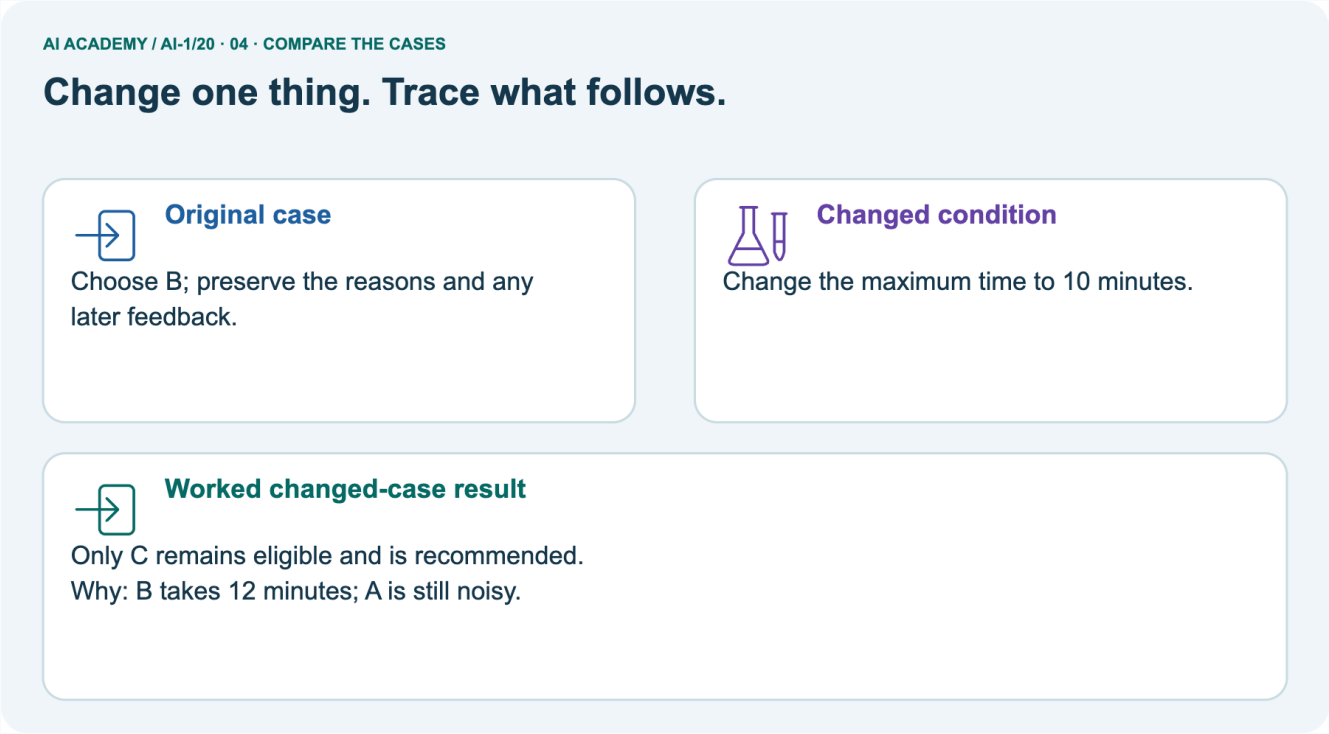


Follow the labelled picture with the matching explanation. Open the vector diagram

Requirements before **preference**s. A: Quiet = No, Minutes = 8, Score = 3, Status = excluded. B: Quiet = Yes, Minutes = 12, Score = 2, Status = eligible. C: Quiet = Yes, Minutes = 10, Score = 1, Status = eligible.

Read the labels and values. Explain the relationship the model shows.

04. Change one thing. Trace what follows.



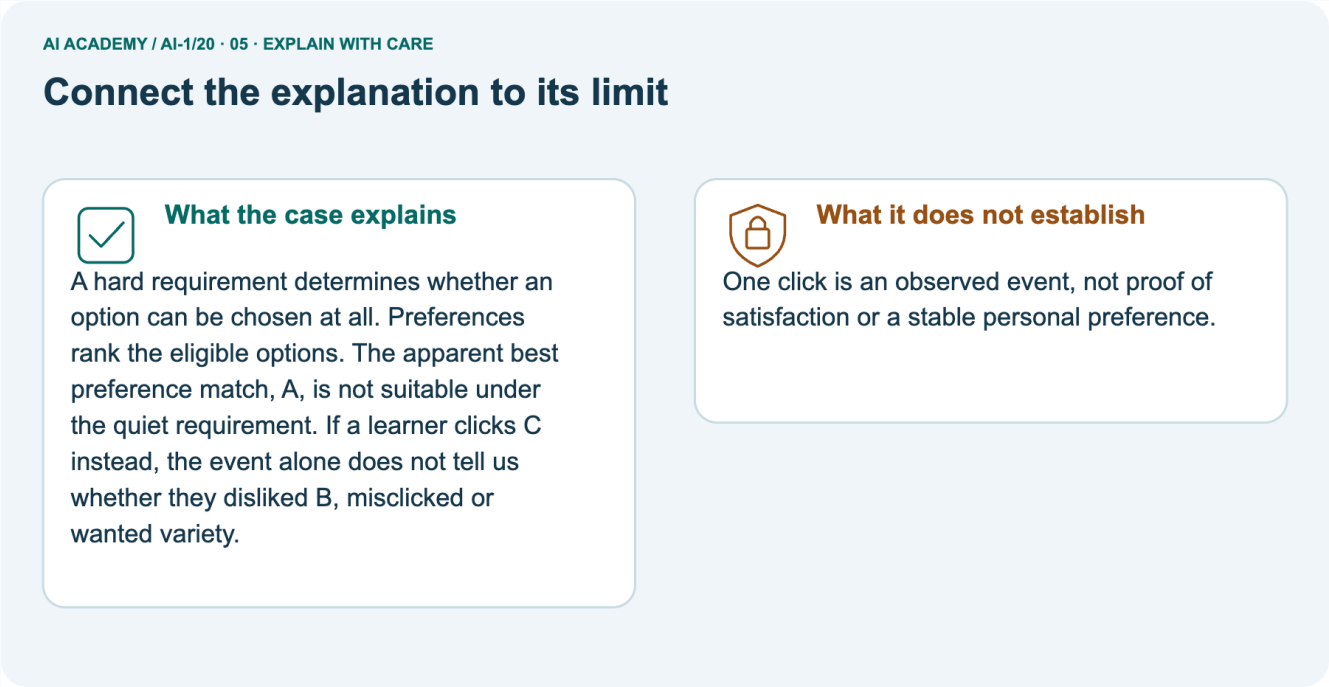
Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: Choose B; preserve the reasons and any later feedback. Changed condition: Change the maximum time to 10 minutes. Expected result: Only C remains eligible and is recommended.

Reason: B takes **12 minutes**; A is still noisy.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

05. Connect the explanation to its limit



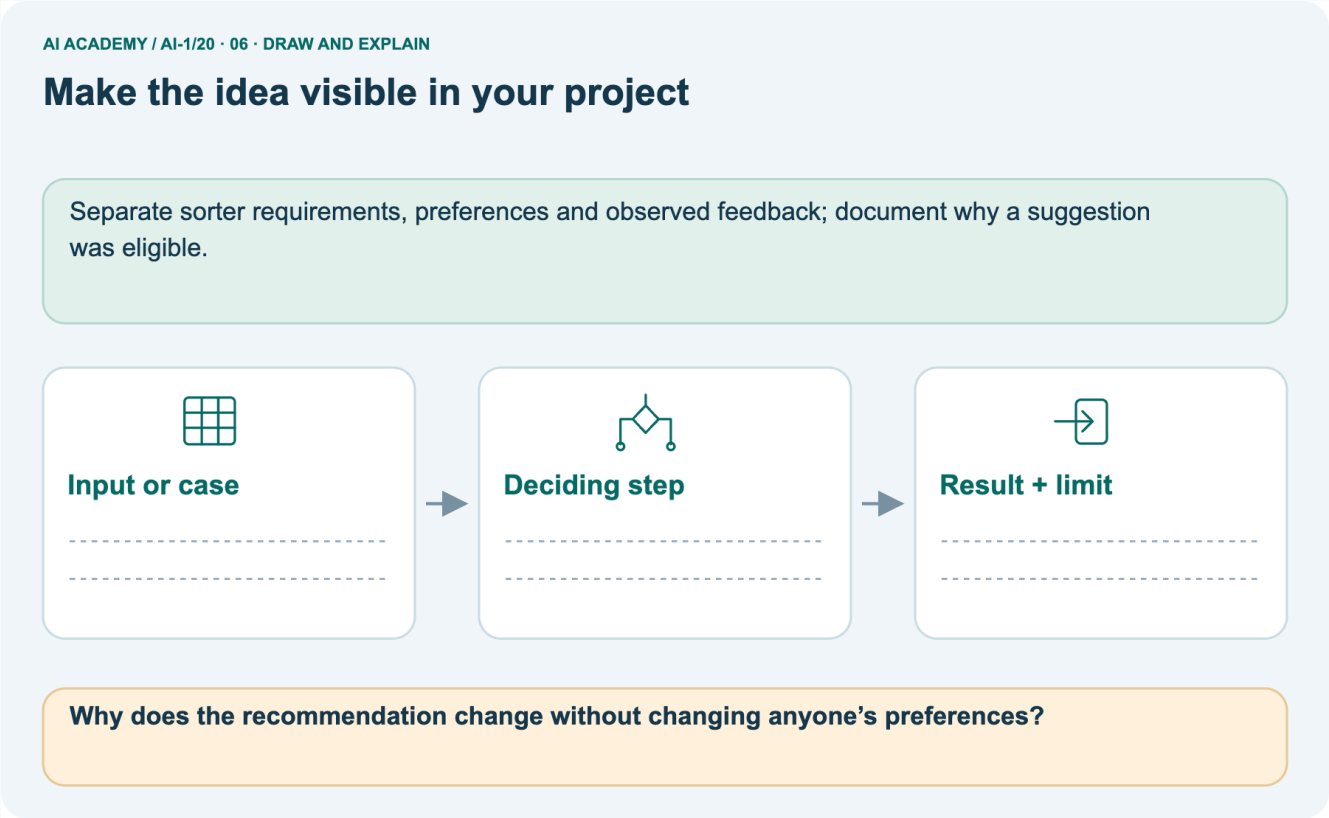
Follow the labelled picture with the matching explanation. Open the vector diagram

A hard requirement determines whether an option can be chosen at all. **Preference**s rank the eligible options. The apparent best **preference** match, A, is not suitable under the quiet requirement.

If a learner clicks C instead, the event alone does not tell us whether they disliked B, misclicked or wanted variety. Limit: One click is an observed event, not proof of satisfaction or a stable personal **preference**.

Explain the case, then point to a limit on the claim.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

Separate sorter requirements, **preference**s and observed feedback; document why a suggestion was eligible. Draw your **input**, deciding step, result and limit.

Supported question: Why does the **recommendation** change without changing anyone’s **preference**s?

Draw your project trace and justify the deciding step using **evidence**.

▼ Earlier wording, alternatives and picture transcripts

01. Recommendations: the case

02. Follow the deciding steps

03. Requirements before preferences

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.